

Introduction

STAT-400

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Objectives

- Understand the complexity of Governmental decisions
- Know the need of using quantitative approach to management decisions
- Appreciate the role of Statistical methods in data analysis.

Role of observations ?

The observation play very important role in all field of study such as:
Agriculture, Industry, Transport, Business, Insurance, Banking and biological sciences etc.

Why study statistics?

1. Data are everywhere
2. Statistical techniques are used to make many decisions that affect our lives
3. No matter what your career, you will make professional decisions that involve data. An understanding of statistical methods will help you make these decisions effectively

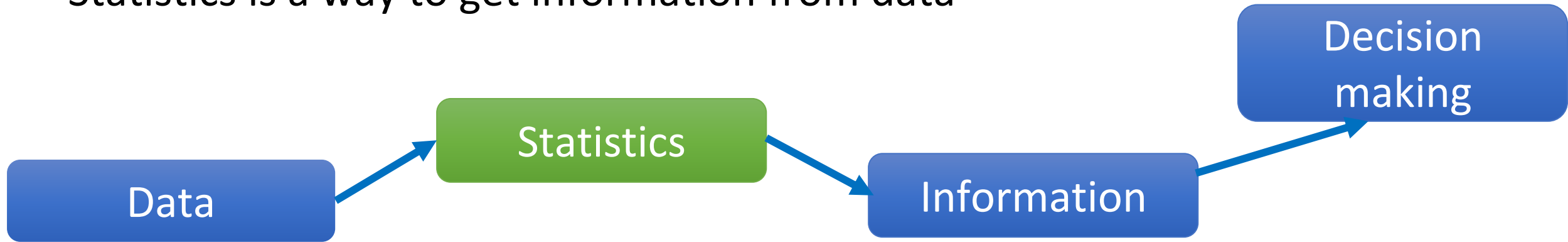
Statistics is a collection of methods for planning experiments, obtaining data, and then processing, summarizing, presenting, analyzing, interpreting, and drawing scientific conclusions based on the data under uncertain conditions.

Variation and Uncertainty

- Statistics is the subject which deals with the variability. No two objects in a universe are exactly alike. If they were, there would have been no statistical problem.
- It also deals with uncertainty as every process of getting observations whether controlled or uncontrolled, involves deficiencies or chance variation. That is why we have to talk in terms of probability since the inferences which are made about the population on the basis of sample evidence cannot be absolutely certain.

What is Statistics?

- “Statistics is a way to get information from data”



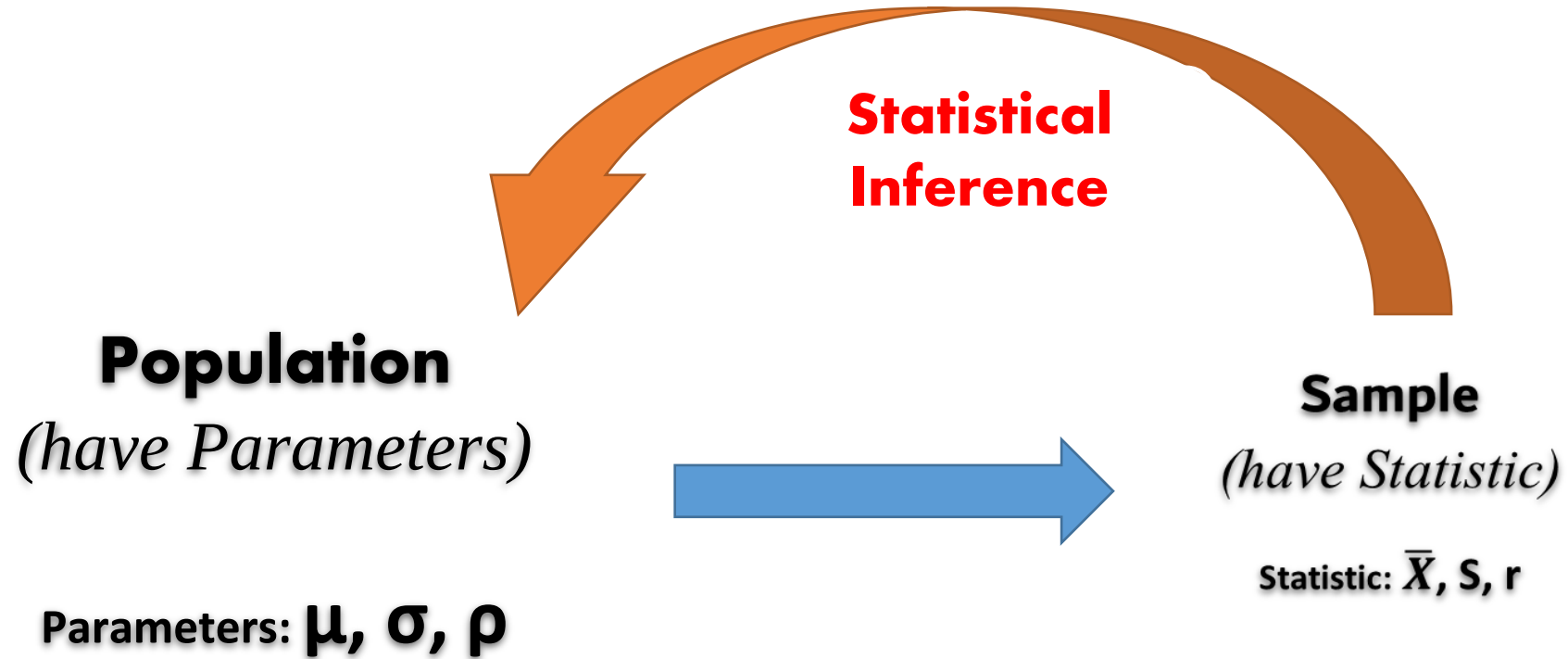
Data: Facts, especially numerical facts, collected together for reference or information.

Information: Communicated concerning some particular facts.

STATISTICS

- **Statistics may be defined as
a science of collection , representation , analysis
and interpretation of numerical data under
uncertainty conditions.**

Population vs Sample



Population: A Population is a group of all object/elements/items under investigation.

Sample: A representative part/subset of the population.

Branches of Statistics

Statistics

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graph TD; Statistics --> Descriptive; Statistics --> Inferential;
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Descriptive

Involves in Organization, Summarization, and Display of Data into Tables, Graphs and Summary Numbers such as $\bar{X}$, S , r , p

Inferential

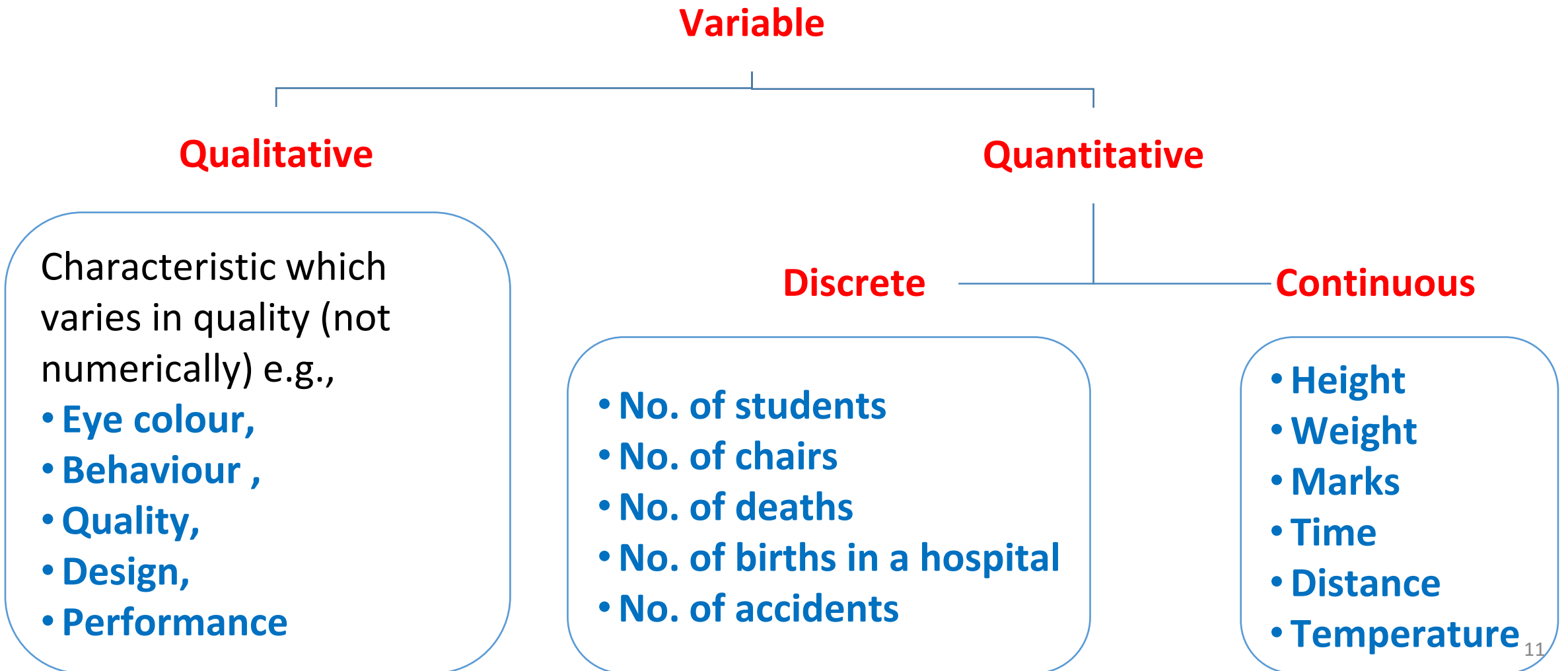
Using sample information such as $\bar{X}$, S , r , p to draw Inference about the Population.

Variable

Any Characteristic that may vary from Object to Object is known as Variable. e.g., marks, age, height, sex, sales, etc.

- Height of a tree
- Number of insects on a tree
- Colour of a flower

Type of Variable



Qualitative Variable

- When the characteristic being studied is nonnumeric, it is called a qualitative variable or an attribute.
- For example, gender, religious affiliation, type of automobile owned, eye colour, etc.
- When the data are qualitative, we are usually interested in how many or what proportion fall in each category.

Quantitative Variable

When the variable studied can be reported numerically, the variable is called a quantitative variable.

For example, balance in your checking account, the ages of company employees, the life of an automobile battery (such as 42 months), and the number of children in a family, etc.

Discrete Variable

Discrete variables can assume only certain values, and there are usually “gaps” between the values.

For example, number of bedrooms in a house (1, 2, 3, 4, etc.)

We count, for example, the number of bedrooms in a house. Notice that a home can have 3 or 4 bedrooms, but it cannot have 3.56 bedrooms. Thus, there is a “gap” between possible values. Typically, discrete variables result from counting.

Continuous Variable

Continuous variable can assume any value within a specific range, i.e., its domain is an interval with all possible values without gaps. The continuous variable flows without a break from one value to the next with no limit to the number of distinct values. Examples of continuous variables are the air pressure in a tire and the weight of a shipment of tomatoes, height of a student, etc. Typically, continuous variables result from measuring.

INTRODUCTION TO STATISTICS

➤ **Data**

The Collection of some related observations is called data. Recording, Picture, Numerical values.

➤ **Classification of data**

Data that may have been originally collected and have not undergone any sort of statistical treatment or statistical method are called ***Primary data***, while the data that have undergone any sort of statistical treatment at least once are called ***Secondary data***.

Data may be available from existing sources e.g. records and publications or the same may have to be collected afresh.

INTRODUCTION TO STATISTICS

➤ **Collection of primary data**

- (1) Direct personal investigation / indirect
- (2) Personal interview.
- (3) Collection through questionnaires.
- (4) Collection through enumerators.
- (5) Collection through local sources

➤ **Collection of Secondary data:**

1. Official Publications

- ☐ Federal/punjab Bureau of Statistics
 - ☐ Population Census(each and every unit of population) of Organization
 - ☐ Ministries of Health, Food, Agriculture, Finance etc. State bank of Pakistan SBP
 - ☐ Provincial Bureaus of Statistics
- (2) Semi-official University, Interloop,

By the number of variables

Univariate data set have single piece of information recorded for each item.
Weight of students

$X = x_1, x_2, x_3, x_4, x_5, x_6, \dots, X_n$

Bivariate data sets have exactly two piece of information recorded for every items.
Dependent or independent

Weight of students, height of students Income and expenditure, Price & demand,
Sale of vehicle and sale of mobile phones , Fertilizer and yield

Multivariate data sets have minimum three piece of information recorded for every items

Weight of students + height of students + income of person

Fertilizer and environment factor = yield

Measurement scales

- Nominal scale
- Ordinal scale



Quality measure: classify into non-numeric form.

- Interval scale
- Ratio scale



Quantitative measure:
Data obtained Numerically.

Nominal Scale:

The nominal scale of measurement uses numbers merely as a measurement separating the objects or events into different classes or categories.

For example:

Gender:

Male / Female

Marital Status:

Single/ Married

Eye colour religion Specialization & Nationality

Ordinal Scale:

The ordinal scale of measurement uses numbers merely as a means of arranging the objects being measured in order. Smallest to highest “or” largest to lowest.

For example:

Student in a class

“Good and Excellent” Average“ below average”

Medical condition of patient

“Satisfactory”, “Serious”, “Guarded”, “Critical”

This kind of measurement is obviously of higher type than that used in obtaining a nominal scale since we are able not only to group individual into separate categories but to order the categories as well.

Interval Scale:

- A variable which measured in an interval scale can be “+”, “-”, “*” but calculating ratio is not possible.
- Zero point in an interval scale is arbitrary. For example, a temperature can be below 0-degree Celsius and or negative.

Ratio scale:

- It is a special kind of measurement where the scale of measurement has a true point as its origin. The ratio scale is used to measure weight, volume, length, distance, money, etc.

“The key to differentiating interval and ratio scale is that the zero point is meaningful for ratio scale.

Some important notations

- Variables are usually denoted by **X**, **Y**, **Z** etc
- Number of values in a data set by **n**
- Sum of all the values of variable **X** $\sum X$
- Sum of squared of all values of **X** $\sum X^2$
- Deviation of values of **X** from **a** $(X-a)$
- Sum of deviation of **X** from **a** $\sum (X - a)$
- Sum of squared deviation of **X** from **a** $\sum (X - a)^2$

Some ingredients of statistics formula

X	X ²
2	4
4	16
5	25
6	36
8	64
25	145

$$(1) \sum X$$

$$(2) \sum X^2$$

$$(3) \left(\sum X \right)^2 = (25)^2 = 625$$

Some ingredients of statistics formula

X	X-6	(X-6)²
2	-4	16
4	-2	4
5	-1	1
6	0	0
8	2	4
25	-5	25

$$(1) \sum (X - 6)$$

$$(2) \sum (X - 6)^2$$

Some ingredients of statistics formula

X	X-5
2	-3
4	-1
5	0
6	1
8	3
25	0

$$(1) \bar{X} = \frac{\sum X}{n} = 5$$

$$(2) \sum (X - \bar{X})$$

Sum of deviations of values from mean is always zero

Some ingredients of statistics formula

X	$(X-5)^2$	$(X-6)^2$
2	9	16
4	1	4
5	0	1
6	1	0
8	9	4
25	20	25

$$(1) \sum (X - \bar{X})^2$$

$$(2) \sum (X - 6)^2$$

Sum of squared deviations
of values from mean is
always minimum

Presentation of data

When the suitable statistical data have been collected, the next step is the condensation or presentation of the data so that valid inferences can be drawn.

Methods for the presentation of data

Presentation of Qualitative data

Example 1: Consider the data about Gender of 10 students

Sex	M	F	M	M	F	M	F	M	M	M
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- Make a frequency distribution, relative frequency and percentage frequency.
- **Example 2: Suppose we have also collected data of Sections of these 10 students as**

Sex	M	F	M	M	F	M	F	M	M	M
Section	A	A	A	B	B	B	A	B	A	B

- Construct the Cross tabulation of the above data and interpret your results?

Sex	f	r.f	% f
Male	7	$7/10=0.7$	$0.7*100=70$
Female	3	$3/10=0.3$	$0.3*100=30$
Total	$\Sigma f=10$	1	100

Sex	Sec A	Sec B	Total
Male	3	4	7
Female	2	1	3
Total	5	5	10

Discrete data – Frequency Distribution

Example:

Following data represent the number of infected plants from a sample of twenty experimental plots. Your task is to present it in tabular form.

1	2	4	3	0	1	2	3	1	1	0
2	1	0	2	3	0	0	1	3		

Range= $X_m - X_0$

X_m = maximum value in the given data set= 4

X_0 =minimum value in the given data set=0

Discrete Frequency Distribution

No. of infected plants X	Tally	Frequency f	Relative Frequency	Percent Frequency
0		5	$5/20=0.25$	$0.25*100=25$
1		6	$6/20=0.30$	$0.30*100=30$
2		4	$4/20=0.20$	$0.20*100=$
3		4	$4/20=0.20$	20
4		1	$1/20=0.05$	$0.05*100=$
Total		20	1	5
				100

Frequency Table for continuous variable

The following data represents the height of 30 wheat plants taken from the experimental area. Construct a frequency distribution.

87	91	89	88	89	91	87	92	90	98	95
97	96	100	101	96	98	99	98	100	102	99
101	105	103	107	105	106	107	112			

Range= (maximum value in data set)-(minimum value in data set)

Following data represents the plant height (cm) of a sample of 30 plants.

87	91	89
88	89	91
87	92	90
98	95	97
96	100	101
96	98	99
98	100	102
99	101	105
103	107	105
106	107	112

Classes	Frequency (f)		
86–90	6		
91–95	4		
96–100	10		
101–105	6		
106–110	3		
111–115	1		
Total	30		

Frequency Distribution

- Tabular arrangement of data in which various items are arranged into classes or groups and the number of items falling in each class is stated.
- The number of observations falling in a particular class is referred to as class frequency "**f**".
- Data presented in the form of a frequency distribution is also called grouped data.

Some definitions

Class Limits

The class limits are defined as the number or the values of the variables which are used to separate two classes. Sometimes classes are taken as

20--25, 20 , 21, 24 25, 27

25—30,

30--35 etc In such a case, these class limits means " 20 but less than 25", "25 but less than 30" etc

Class marks or midpoints

The class mark or the midpoint is that value which divides a class into two equal parts. It is obtained by dividing the sum of lower and upper class limits or class boundaries of a class by 2.

Class interval The difference between either two successive lower class limits or two successive upper class limits or two successive midpoints and denoted by "h".

Construction of a frequency distribution

- Decide the number of classes:

$$K=1+3.3 \log(n)=5.87 \text{ or } \sqrt{n}=5.47 \rightarrow 6 \text{ Classes}$$

- Determine the range of variation of the data i.e,

$$R= \text{Max} - \text{Min} = 112 - 87 = 25$$

- Determine the approximate size of class interval

$$h = \frac{R}{K} = 25/6 = 4.17 \rightarrow 5 \text{ Class Interval}$$

- Decide where to locate the class limit of first class

Just below the minimum value in the data $\rightarrow$ 86-90, 91-95, ...

- Distribute the data into appropriate classes

Use Tally (|) for each value

Frequency Distribution

Classes	Class Boundaries	Tally	Freq (f)	Mid-Point (X)	Class Interval (h)	c.f.	r.f.	% freq
86–90	85.5–90.5		6	88	5	6	0.200	20.0
91–95	90.5–95.5		4	93	5	10	0.133	13.3
96–100	95.5–100.5		10	98	5	20	0.333	33.3
101–105	100.5–105.5		6	103	5	26	0.200	20.0
106–110	105.5–110.5		3	108	5	29	0.100	10.0
111–115	110.5–115.5		1	113	5	30	0.033	3.30
Total			30				1.000	100.0